

# Data Cleaning Report

## Student Success Analytics Project

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### Introduction

This report documents the data cleaning process performed on a University's student dataset that contains 315 records and 9 variables. The objective was to prepare the data for exploratory analysis into which factors are strongly associated with student academic success. Data cleaning was necessary because missing values, duplicate records, and inconsistent formatting can reduce data quality and lead to inaccurate analytical results.

### Dataset Overview

| Item                | Value                       |
|---------------------|-----------------------------|
| Number of Records   | 315                         |
| Number of Variables | 9                           |
| Source              | AI-generated sample dataset |
| File Type           | CSV                         |

### Tools Used

| Tool                   | Purpose                                                                 |
|------------------------|-------------------------------------------------------------------------|
| Excel                  | Initial data inspection                                                 |
| Python (Pandas, NumPy) | Missing value inspection, duplicate value inspection, and data cleaning |
| SQL                    | Data analysis                                                           |
| Tableau                | Data visualization                                                      |

### Issues Identified

- Missing values
  - Found 57 missing values in ParentEducation
  - Found 72 missing values in InternetAccess
  - Found 20 missing values in StudyHoursPerWeek
  - Found 20 missing values in FinalGrade
- Duplicate rows
  - Found 12 duplicate rows
- Standardization

- The gender column had values that were written in all capital letters or all lowercase letters.
- The InternetAccess column had values that were written as “Y” or “N”.

### **Data Cleaning Procedures**

- Missing values were replaced using the `df.fillna` function in Python.
  - Missing values in ParentEducation were replaced with “Unknown” to indicate that no record was made for the education of the parent.
  - Missing values in InternetAccess were replaced with “Unknown” to indicate that no record was made for the student's internet access of the student.
  - Missing values in StudyHoursPerWeek were replaced with the median of the values, as the median is less sensitive to extreme values and helps preserve the overall distribution of the data.
  - Missing values in FinalGrade were removed because FinalGrade is the outcome that is being analyzed. Giving missing values a 0 or some other value could prevent accurate evaluation of student success.
- Duplicate rows were removed because they could artificially increase counts and bias statistical summaries and visualizations.
- Standardization was performed to ensure consistent category labels across records, allowing accurate grouping, filtering, and reporting during analysis.
  - The values in the gender column now read “Male” or “Female”
  - The values in the InternetAccess column now read “Yes”, “No”, or “Unknown”

### **Summary of Changes**

- There was a total of 169 missing values:
  - 57 in ParentEducation
  - 72 in InternetAccess
  - 20 in StudyHoursPerWeek
  - 20 in FinalGrade
- Missing values were replaced with the word “Unknown”, the median, or removed altogether to improve consistency and give accurate insights.
- Duplicated rows were removed as they could have led to inaccurate insights.
- Gender and InternetAccess were standardized to create consistent reporting.
- The cleaned dataset now has 283 records and 9 variables, compared to the 315 records and 9 variables before the dataset was cleaned.

### **Limitations**

While these changes made a consistent dataset, there could be some limitations to the changes that were made. For instance, having an “Unknown” label in ParentEducation and InternetAccess does not mean that the parent does not have an education history or that the student does not have access to the internet. Similarly, having the median in StudyHoursPerWeek does not mean that this was the student's true study hours per week. The fact that the FinalGrade rows were removed does not mean that the student did not receive a final grade.

## **Conclusion**

- The dataset was successfully cleaned and prepared for analysis.
- Missing values were identified and replaced or deleted.
- Duplicate values were removed.
- The resulting dataset contains 283 complete and consistent observations suitable for further analysis.